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Evolution of business physiology in the wine industry: Insights from the Stra.Tech.Man Scorecard in the Cephalonian Robola sector

This study explores the physiological evolution of entrepreneurship in the Robola PDO wine sector in Cephalonia, Greece, utilizing the Stra.Tech.Man Scorecard as a diagnostic tool. Focusing on micro-firms producing and marketing the Robola PDO wine variety, the research provides insights into the executives' self-assessments of their strategic, technological, and management adaptation efforts from 2017 to 2021. The findings indicate that these firms exhibit a strong strategic ambition to improve their competitiveness, but encounter obstacles in technology and networking operations, as well as limitations in developing their human resources and their overall management methodologies. Furthermore, the study reveals that the companies' innovation potential was enhanced during the COVID-19 pandemic. The study highlights the potential for developing a sophisticated regional wine brand and underscores the role of innovative entrepreneurs and government support in fostering a competitive and sustainable entrepreneurial ecosystem.

Keywords: business physiology; Stra.Tech.Man Scorecard; Cephalonian Robola

1. Introduction

In 2004, Ch. Vlados conducted a seminal study examining the evolutionary dynamics of strategy, technology, and management in Greek businesses, particularly within the food and beverage industry during the 1990s (Vlados, 2004). This research introduced the concept of "business physiology," which employed the Stra.Tech.Man synthesis (strategy-technology-management) to view companies as living socioeconomic entities with a dialectic, structural, and evolutionary nature. Drawing from evolutionary economics and biological analogies (Hodgson, 1993; Nelson & Winter, 1982), this approach argues that organizations typically undergo gradual physiological changes rather than abrupt ones, except in instances of rapid hybridization. It highlights the organic processes that drive business transformation due to evolutionary complexity. The

Stra.Tech.Man innovation triangle, a dialectical synthesis of strategy, technology, and management, is developed as an adaptation function, enabling socioeconomic organizations, particularly firms, to competitively adapt and evolve in the ever-changing landscape of globalization. Later research has explored the applicability of this approach across various sectors, regions, and entrepreneurial dynamics, while also acknowledging its limitations (Chatzinikolaou & Vlados, 2022; Falaras & Moschidis, 2021; Vlados & Chatzinikolaou, 2019).

Efforts continue to rigorously evaluate this approach, as its implementation in various local business ecosystems is constrained by typically non-representative sample sizes and limited capacity for replication and expansion. Acknowledging the current literature's boundaries, this study seeks to augment it by investigating the physiological transformation of firms and the evolution of entrepreneurship in the Greek wine sector, emphasizing the Protected Designation of Origin (PDO) variety of Robola cultivated on the island of Cephalonia. This research aspires to offer valuable insights and enhance comprehension of the topic, considering the existing limitations.

This research endeavors to illuminate the driving forces behind the physiological evolution of entrepreneurship in Cephalonia's wine industry, specifically in relation to the PDO Robola variety. Furthermore, it investigates the ways in which environmental shifts influence the Stra.Tech.Man physiology of the firms being studied.

The importance of this analysis stems from the potential to enhance our comprehension of the Stra.Tech.Man approach's impact on entrepreneurial dynamics and business physiology across various sectors and regions. By delving into these questions, the study aspires to generate novel insights that may guide future predictions and developments within the field.

The structure of this paper unfolds as follows. Section 2 presents a comprehensive review of the literature, delving into the Stra.Tech.Man approach and its application to strategic audits, followed by an exploration of the Greek wine industry with a focus on the PDO Robola variety. In Section 3, the paper outlines its methodological approach, presenting the Stra.Tech.Man Scorecard and research identity, along with a description of the demographic characteristics of the sample. Sections 4 and 5 reveal the empirical findings and engage in a nuanced discussion of respondents' self-assessments, their interpretations, and the implications for the regional wine brand's trajectory. Lastly, Section 6 reflects on the limitations of the study and delineates prospects for future inquiries, addressing both practical and theoretical ramifications.

2. Literature review

2.1. Strategic audit and the Stra.Tech.Man approach

Business strategy evaluation is an essential component of entrepreneurship literature. According to fundamental approaches by Donaldson (1995), Hunger and Wheelen (1996), Gordon (1997), Grundy (2008), and Seifzadeh and Rowe (2019), strategic auditing is a diagnostic mechanism that aims to identify problem areas and strengths of an organization by evaluating implemented strategies based on measured performance.

An important extension of this theme is the Balanced Scorecard (BSC) introduced by Kaplan and Norton (1992). Kaplan and Norton's BSC is widely recognized as a strategic control mechanism in both academic and business communities. It functions as a cockpit, helping to monitor and measure the performance of various functions within an enterprise. Kaplan and Norton's BSC records and analyzes a company's objectives and measures from four key perspectives: financial, customer, internal environment, and

innovation. Although Kaplan and Norton (2001, 2004) made some modifications to the original model, the central assumptions remained unchanged.

The literature on Kaplan and Norton's BSC contains various criticisms, the main one being that it seems to deviate from the fundamental assumptions of the original "tableau du bord." Critics accuse Kaplan and Norton's BSC of treating firms quasi-mechanically, being internally focused exclusively and rigid (e.g., Kaplan & Norton, 2006; Voelpel et al., 2006).

Vlados (2021) argues that the conventional BSC approach is generally more appropriate for larger, sophisticated companies with well-developed performance measurement systems. These companies are capable of handling the extensive data required, which may be difficult for smaller and medium-sized enterprises (SMEs) to process. Instead, the author proposes a business physiology approach that focuses on a company's responses to key evolutionary questions in the areas of strategy, technology, and management through a periodic and longitudinal self-evaluation process. These responses contribute to the development of a company's physiological innovative profile and dynamics.

As illustrated in Figure 1, a company's strategy revolves around fundamental questions such as "Where am I and where do I want to go? How do I get there? Why?" The technology aspect addresses questions like "How do I derive, create, synthesize, disseminate, and reproduce the means of my work and knowledge? Why?" Finally, management pertains to questions such as "How do I use my available resources? Why?" A company's responses to these questions define its innovative synthesis, signifying its endeavors to adapt effectively to shifts in the external environment, thereby guiding the development of its distinct business physiology.

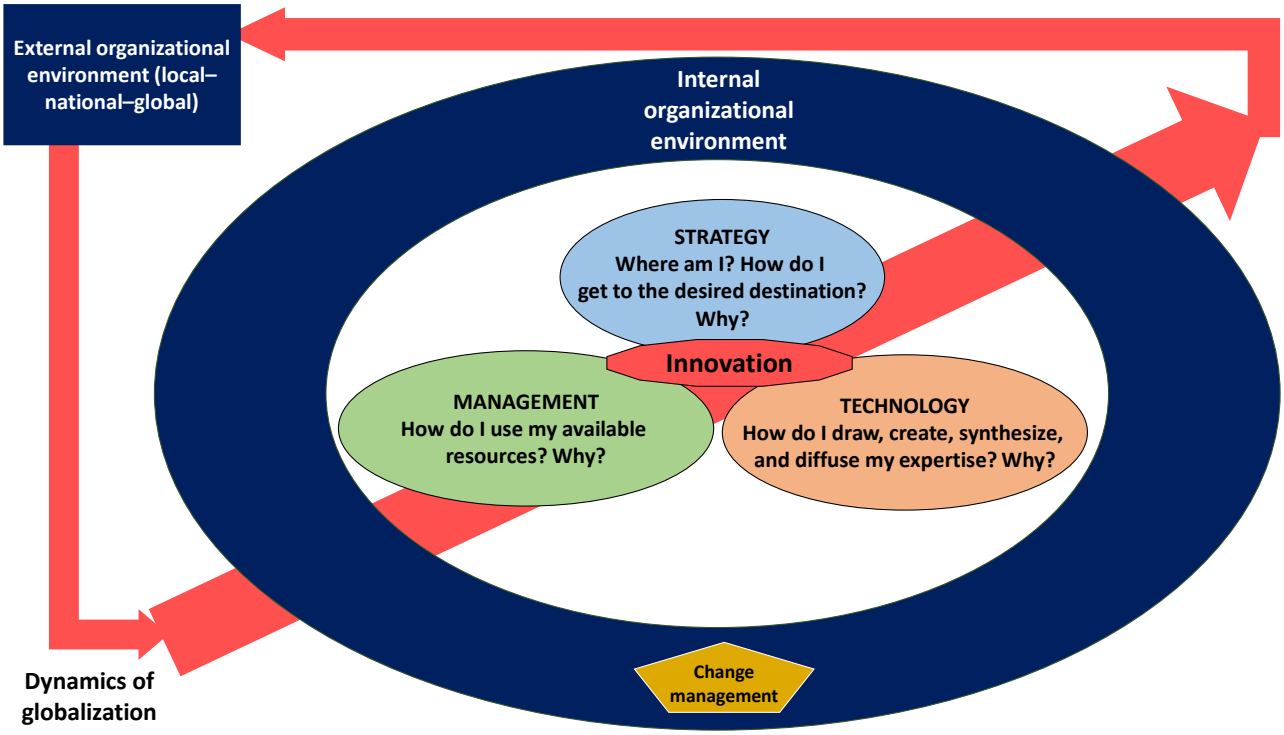


Figure 1: The Stra.Tech.Man approach. Based on Vlados (2004)

Three primary entrepreneurial physiologies emerge from the synthesis of evolutionary strategy, technology, and management (Vlados, 2004): monad-centric, massive, and flexible. Monad-centric businesses largely rely on instinctive strategic decisions, sporadic technological choices, and management rooted in daily practice. This category encompasses a significant portion of family-owned SMEs in Greece across various sectors (Vassiliadis & Vassiliadis, 2014). Massive businesses are characterized by mechanistic strategic approaches, linear technology accumulation, and management focused on job specialization. In Greece, these typically include larger companies that exploit economies of scale and extensive sales networks (Binda, 2013). Flexible business physiologies involve a dialectical mix of innovative and disruptive strategic designs, progressively networked advanced technology, and open-minded, inclusive management that encourages participatory forms (Sushil et al., 2015). According to this perspective, the Greek business ecosystem primarily comprises monad-centered firms (Vlados et al.,

2021). The structural features and specificities of each socioeconomic environment, including Greece, appear to facilitate the transformation of business physiologies, thereby contributing, to some extent, to a persistent lag in the competitiveness of the Greek economy (Chatzinikolaou & Vlados, 2022).

Consequently, the strategic audit and the Stra.Tech.Man approach offer invaluable insights for assessing and enhancing business performance. Although Kaplan and Norton's BSC has been extensively employed and recognized as a strategic control mechanism, it presents limitations, especially for smaller organizations like SMEs. The Stra.Tech.Man approach delivers a more accessible and comprehensive self-assessment model that tackles the evolving dimensions of strategy, technology, and management. Given the upcoming discussion on the Greek wine industry, it is highly pertinent to examine how this comprehensive framework can be employed to assess and enhance the sector's performance.

2.2. The wine industry in Greece

Several relevant studies can be found in the literature concerning the wine industry. Sampedro et al. (2010) introduced the environmental aspect as a critical success factor in the management control mechanism of wine production companies. Essentially, they proposed a new type of BSC. Additionally, Oyaneder and Valderrama (2016) examined the performance management systems of specific companies in the wine industry. They aimed to identify where they mainly focus and whether they use structured and sophisticated Kaplan and Norton's BSC-type approaches. The study found that these firms only use BSC-type variations to a small extent. This limited use seems to reduce the potential applied utility of this type of strategic control. Furthermore, Falaras and Moschidis (2021) conducted research to validate the Stra.Tech.Man model in the Greek

wine industry. Their study concluded that this model can help distinguish between functional and more profound organizational innovations in the studied sector.

The alcoholic beverage sector holds a pivotal position in the Greek economy, significantly influencing GDP, employment, and tax revenue (Foundation for Economic & Industrial Research, 2022). Specifically, Greek wine production experienced an 8.13% growth in 2021, reaching a total of 2,468,945 kiloliters, as reported by the National Agricultural Cooperative of Vine and Wine Products (keosoe.gr). A substantial 65.93% of this output comprised wines without geographical indication, while protected geographical indication wines represented 16.84% and varietal wines made up 8.9%. Figure 2 below encapsulates the primary aggregate economic indicators of this industry.

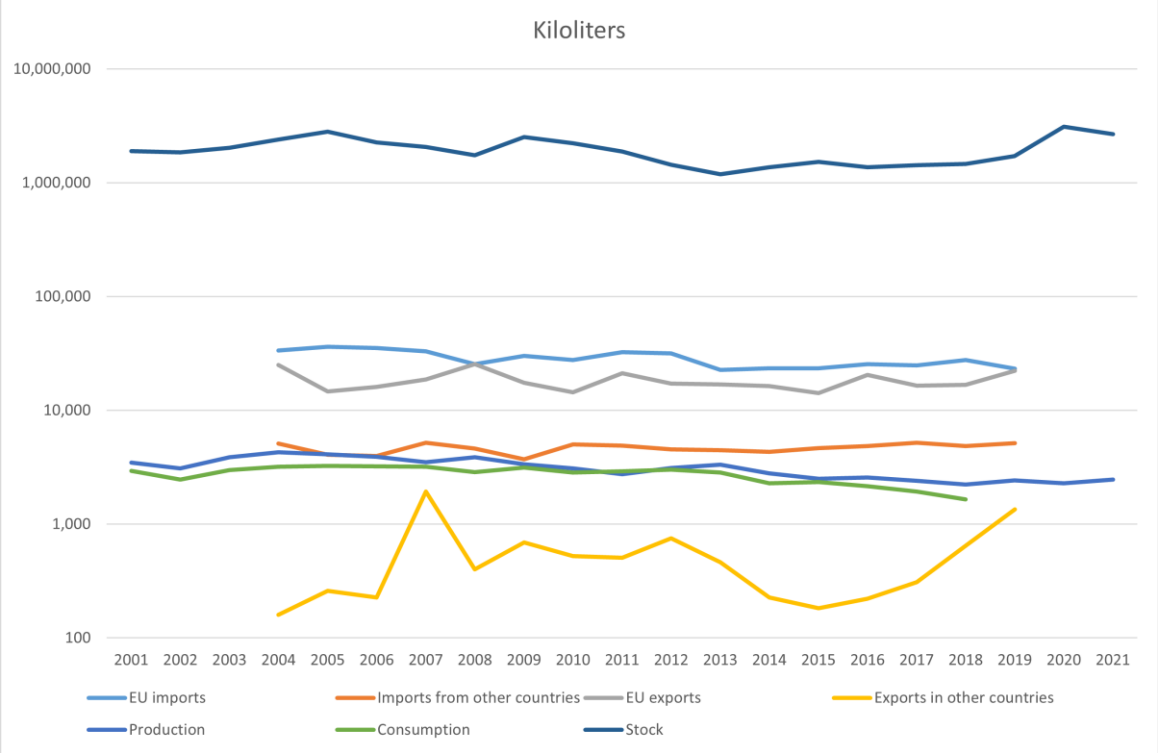


Figure 2: Data of the Greek wine sector over time. Source: National Agricultural Cooperative of Vine and Wine Products (keosoe.gr)

Nevertheless, wine production in Greece shows a cumulative decrease of 30% compared to 2001. Considering 2005, when production was at its highest, this fall accounts for 70%.

The stock of wine has remained relatively stable, although the recent pandemic crisis has led to a larger-than-expected build-up as consumption has fallen sharply. The available data show a significant decline long before the pandemic. It is likely that the prolonged economic crisis significantly impacted consumption habits, and the wine market has not yet recovered from this shock. In particular, consumption experienced a substantial drop after 2012, which continued until 2018. Additionally, there was a decline in imports from the EU after 2012, with a slight upward trend in the following years, while imports from third countries remain stable, accounting for around 15-20% of imports from the EU. Finally, exports are smaller in quantity compared to imports and mainly occur in EU countries. It is worth noting that in 2019, exports to the EU were at least 15 times more than to third countries. The recent increase in exports is likely due to the gradual recovery of the Greek economy after the profound socioeconomic and institutional crisis in 2008 (Andreou et al., 2017; Vlados et al., 2021).

Recent studies have illuminated certain aspects of the Greek wine industry, highlighting its modest comparative performance. Vlachos (2017) posits that Greece, unlike other Southern European Union nations with a history of wine production, has not yet fully capitalized on the sector's export potential. Nonetheless, with appropriate policy measures, a competitive industry featuring globally recognized labels could emerge. Vlachos (2017) further contends that international trade has a greater impact on Greek wine production than domestic consumption, and the high cost of securing qualified personnel poses a significant challenge. In conclusion, it appears that the Greek wine industry is making a significant contribution to the nation's economy in terms of GDP and employment; however, it must address challenges associated with global competitiveness and export potential.

2.3. The PDO Robola variety

Regarding the distribution of vine cultivation in Greece, the Ionian Islands perform well above the average (Cephalonia is the largest island in the region). According to the recent agriculture and livestock census of the Hellenic Statistical Authority (2022), the share of vines in the agricultural production of the Ionian Islands region is 4%, while the regional average is 2.5%. Robola covers 1,500 acres of vineyards, several of which are organically grown. Another 1,500 hectares are planted with Tsaousi, Vostilidi, Mavrodaphne (PDO), Moschatos of Cephalonia (PDO), and Zakynthino. Currently, there are 15 wineries in Cephalonia producing Robola (Greek Gastronomy Guide, 2022). These businesses operate mostly autonomously, although an association aims at joint promotion with a view to their further development.

Regarding the characteristics that make this PDO variety stand out, its price first deserves mention. These are some of the most expensive grapes in Greece, with Cephalonia's wineries paying up to €0.80 per kilo to their producers. Robola is best when grown at high altitudes (above 300 meters) on sloping, well-drained, gravelly soils that are poor in organic matter (Lazarakis, 2005). In Cephalonia, Robola vines are often ungrafted in the limestone soils of this island. The vines are early maturing and can produce wines with high acidity and significant levels of phenolics. Wines from this grape are dry, medium-bodied wines with solid lemon notes. The vine is vigorous but sensitive to water stress and susceptible to various associated viruses (Robinson, 1996). The bunches are relatively small and loose, producing small berries. Robola is compatible with most rootstocks currently available in Greece but is still primarily an ungrafted variety, grown as a bush vine (Lazarakis, 2005). Robola is susceptible to oxidation in the cellar, and delicate handling is required from harvesting the fruit to pressing. The best results are obtained using the latest technology: small pneumatic presses and whole bunch

pressing. Some growers pick the grapes too early because they fear losing their acidity. However, some examples of higher strength (13.2% alcohol by volume or more) show that higher grape maturity can give Robola extra depth and complexity (Directorate-General for Agriculture and Rural Development, 1981).

In summary, the PDO Robola variety appears to be a distinctive and valuable component of the Greek wine industry, which has demonstrated significant growth over the past years. This unique variety possesses specific characteristics that differentiate it from other types. The cultivation and production of Robola wines necessitate specialized knowledge and expertise, as well as meticulous handling throughout the process. As the analysis progresses towards the methodology and application of the Stra.Tech.Man Scorecard, it is essential to consider how the strategic approaches of Kaplan and Norton's BSC-type, as discussed thus far, can be utilized to assess and improve the performance of the PDO Robola variety, ultimately contributing to the overall competitiveness of the Greek wine industry.

3. Method

3.1. The Stra.Tech.Man Scorecard

Drawing on the Stra.Tech.Man approach as its theoretical foundation, the associated Scorecard methodology proposes a comprehensive annual self-evaluation of the company. This assessment, blending quantitative and qualitative aspects, involves the active participation of all senior management members. Its objective is to ascertain the impact of the innovative evolutionary synthesis, as it pertains to the Stra.Tech.Man dimensions, on the company's financial performance. The synthesis of strategy, technology, and management questions, as depicted in Figure 2, serves as the foundation for the Stra.Tech.Man Scorecard.

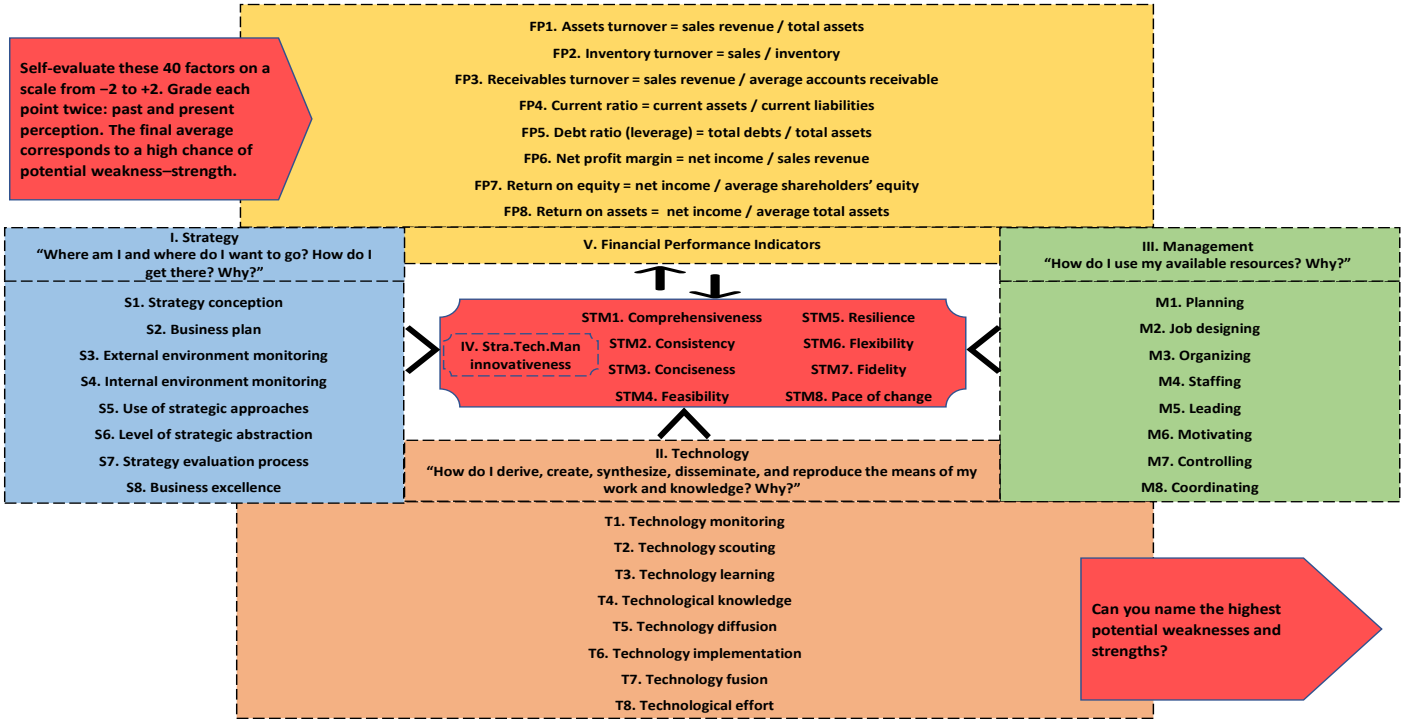


Figure 2: The Stra.Tech.Man Scorecard approach. Based on Vlados (2021)

The diagnostic questionnaires consist of 40 co-evaluated criteria related to the firm's evolution. Respondents are required to fill in their past and present perceptions (from pre-COVID-19 until today, spanning 2017 to 2021) using self-assessment and action research methods (Lewin, 1946). They must score the 40 criteria on a scale from -2 to +2, with the score indicating a high probability of a comparative weak or strong point (Vlados, 2019). The questionnaire also includes an open-ended response section for respondents to justify their scores. Senior management should interpret the averages and open-ended responses collectively, identifying weak points and implementing corrective action through in-house consultation —a step proposed to be conducted in collaboration with specialized researchers or consultants. Specifically, the Stra.Tech.Man Scorecard organizes the self-assessment into five domains. The researcher then synthesizes these domains so that stakeholders can coherently understand their results. The ultimate goal is to transform the business into new, more effective forms and trajectories of Stra.Tech.Man physiology.

Vlados (2021) posits that any organization can employ this diagnostic method, regardless of whether they have the means to consistently track quantitative data beyond essential financial figures and performance indicators. This approach allows managers to conduct self-assessments of their company's performance, setting it apart from Kaplan and Norton's (1992) original BSC. Moreover, Vlados (2021) recommends investigating the evolutionary potential of larger sectoral and spatial clusters, provided that researchers acquire data from representative samples. Therefore, the Stra.Tech.Man approach was chosen as the data collection method in this study due to its comprehensive evaluation of the firm's evolution.

3.2. Research identity

The study focused on five wineries producing the Robola variety in the prefecture of Cephalonia. A Stra.Tech.Man Scorecard questionnaire was distributed to the senior management of these companies. Subsequently, in-depth interviews were conducted in the first half of 2022 (see also Repousis, 2022) through business visits and systematic action research with frequent communication (Wilson, 2021). Four of the six wineries of the Cephalonia Winemakers Association and one outside this organization participated —the five wineries together account for about 80% of the bottled wine production in Cephalonia, which provides sufficient coverage in the representativeness of this survey. Therefore, a form of convenience —and to some extent purposive— sampling was followed to reach firms willing to participate in the survey (Lavrakas, 2008).

In Table 1, demographic data of the examined cases are presented. The abbreviation "Win" is used when referring to one of the five cases because all respondents were promised that personal or confidential information would not be published.[1]

	Win1	**Win2**	**Win3**	**Win4**	**Win5**
Number of employees	5	10	10	6	5
Distribution areas (local, regional, national, international)	80% local, 20% regional	40% local, 20% national, 40% international	80% local, 10% national, 10% international	10% local, 30% national, 60% international	40% local, 30% national, 40% international

Table 1. The sample.

The enterprises studied are micro-firms, as the number of employees does not exceed 10. Nevertheless, their export activity appears relatively dynamic, and it is expected that these firms will be resilient and possess significant growth potential. Two were founded in the early 1980s when Robola became a PDO (Directorate-General for Agriculture and Rural Development, 1981). The other three were established more recently, in the 2010s.

This study adopted a mixed-method approach, incorporating both quantitative and qualitative components. A five-point Likert scale, with scores ranging from -2 to 2, was employed for qualitative data analysis. Responses from the 40-question survey were recorded and averaged, both per company and for the entire sample (for the full questionnaire, see Vlados, 2021). Data collected assessed perceptions and attitudes from 2017 to 2021, covering the period before and during the COVID-19 pandemic. Financial statements for three out of five companies were obtained and analyzed to determine the Stra.Tech.Man Scorecard ratios, which were then compared with self-assessment outcomes. For the remaining two companies, additional information was requested from

[1] All raw data are stored in spreadsheets, which are available to interested researchers upon request.

respondents due to the unavailability of published statements. Similarly, Winery 3 lacked public statements after 2020, and further clarifications were sought.

The theory suggests concurrently using qualitative questions to interpret the elements of subjectivity and emotional intensity inherent in scaled self-assessments (Batterton & Hale, 2017; Harpe, 2015). This diagnostic method allows respondents to provide in-depth information about the issues under consideration (Worley, 2015). Thus, the study examined the qualitative responses received from the open-ended justification part of the 40 criteria (up to 100 words). The responses to the final question, which was a broader discussion of strengths and weaknesses, with a range of up to 500 words, were also analyzed.

When the questionnaire is structured and the samples are relatively small in these mixed surveys, broader interaction with respondents can help interpret the results (Saunders et al., 2019). The findings of this mixed method are mainly qualitative, analyzing the meanings given by the research subjects themselves through their perspective and everyday experience —quantitative data help to provide a holistic understanding of phenomena (Creswell, 2014). Thus, this mixed method helps to avoid bias in responses as the researcher's presence may lead respondents to "embellish" the situation and hide elements of reality. This problem can also be addressed by an in-depth description of the interaction between researcher and respondent, which can be analyzed from a historical perspective —following Ryle's (2009) approach toward this in-depth qualitative interpretation.

4. Results

The self-assessments all follow the same pattern. As illustrated in Figure 4, perceived improvement is evident for all the companies examined.



Figure 4: Average self-assessments per company

This "star" shape in Figure 4 indicates that these firms display remarkable homogeneity in their perceptions. Figure 5 presents the results in detail.



Figure 5: Overall averages per question

These companies seem to exhibit shortcomings in their strategic analysis of the internal environment. Specifically, their performance in strategic abstraction dimensions, such as vision, mission, and philosophy, is noteworthy, as high scores are recorded for both time periods (pre-COVID-19 and present). Open-ended questions reveal that the general understanding of strategy mainly pertains to the desired outward focus, with the exception of Winery 1, which concentrates on enhancing the product's technical features. Regarding

strategic approaches, challenges arise in adopting various perspectives. Participating in the Cephalonia winemakers' association and conducting market research in comparison to competitors constitute strategic actions rather than approaches. The comprehension and development of the company's vision is also significant, with an emphasis on continuous enhancement. Furthermore, respondents cite their experimentation with new products as a pursuit of excellence. Consequently, Winery 1 displays a dedication to improving quality, not just in quantitative terms. The other wineries frequently mention strategic outward-focused initiatives they are undertaking, such as participation in international exhibitions. On the whole, the strategy of these companies seems to incorporate elements of an evolutionary approach without being confined to short-term and occasional decisions.

In technology, perceived improvement in the self-assessment conducted by respondents is also observed. The score on technological knowledge is remarkable, with entrepreneurs reporting that they are skilled in their field, effectively targeting continuous learning. At first glance, the score on technology fusion is concerning. However, an analysis of the responses indicates that these enterprises are not systematically exploiting hybrid combinations, striving to use state-of-the-art methods. Nevertheless, Robola is the result of a traditional cultivation method. As Winery 3's technology affairs representative states, their production is based on local tradition and land exploitation that has remained unchanged for decades. Also, in more specific technological dimensions, recent procurement of new equipment and weather stations, combined with investments in process automation, is observed. Additionally, they mention cooperation with external consultants and learning through seminars on technology scouting. From a generic perspective, these companies seem to take significant advantage of their competitive position and are leaders in this market niche.

Regarding the management factors examined, deficiencies in organization and leadership are observed, which are characteristics of most micro-firms. On the positive side, coordination scores high, suggesting that these firms no longer function in a confined, monad-centric manner. All the wineries seemingly attempt to establish systematic planning processes and job designing mechanisms, albeit not so successfully. According to some respondents, the division of labor appears problematic due to overlapping tasks. Additionally, Winery 5 refers to being at about 30% of the planning, indicating a specific and quantified plan implemented and monitored at regular intervals. Management control is another aspect where comparative deficiencies are found, as responses are limited to product quality dimensions. Only Winery 1 refers to monitoring a control plan; however, it is the only company in the sample without export activity. From an overarching point of view, these firms are primarily characterized by increased inclusiveness, as the core of employees consists of skilled managers.

In terms of innovation dimensions, improvements are also observed at all levels. Innovative comprehensiveness seems to be an exception, as firms seem to be at a different point than desired. According to the respondents, comprehensiveness is not achieved to the desired degree due to the uncertainty generated by this business model based on a sensitive agricultural crop. More specifically, the manager of Winery 1 argues that the company cannot control all the factors affecting production, while two other wineries emphasize the role of consultants in achieving comprehensiveness. Consistency is probably achieved due to the wineries' small size and the barriers to entry in the sector. Conciseness appears to be covered by the high-level training of the employees; likewise, feasibility by using the appropriate tools to carry out all the necessary tasks. Resilience arises as survival after the pandemic and flexibility from the ability of managers to adapt. Winery 3 is an exception, as it seems to be characterized by a particular organizational

structure that hampers the ability of managers to make the necessary changes on time. Concerning fidelity, the commitment of most respondents to implement feasible innovations, such as the adoption of organic vine cultivation, is observed. Finally, the pace of change is mainly assessed through the external factor of the pandemic, to which all respondents consider that they responded partially satisfactorily.

In conclusion, respondents perceive a noticeable improvement in financial indicators, as illustrated in Figure 6. This upward trend in self-assessments reveals that entrepreneurs across all financial ratios experienced positive growth, attributed to various factors.



Figure 6: Comparative analysis of financial performance indicators: Results from self-assessments

Wineries 1-3 saw increased assets turnover, inventory turnover, receivables turnover, and return on assets, implying more efficient resource utilization and enhanced management of inventory and receivables. Wineries 4 and 5 also improved in these areas, albeit to a lesser degree. The current ratio, which measures liquidity, improved for all wineries,

suggesting a better ability to meet short-term obligations. Notably, Winery 4's debt ratio (leverage) rose substantially from -1 to 2, indicating a higher debt-to-equity ratio, potentially providing additional funds for investment and growth. All wineries reported improvements in net profit margin, return on equity, and return on assets, signifying increased profitability and efficiency in generating returns from investments. These enhancements are primarily attributed to cost-saving measures, improved pricing strategies, and a post-COVID-19 surge in product demand. In summary, the entrepreneurs of these small wineries have observed an increase in their financial ratios, which may be due to enhanced strategic planning, technological operations and superior resource management, as well as a heightened emphasis on competitiveness and profitability.

5. Discussion

This section begins by comparing the strengths and weaknesses of the selected Cephalonian wineries, representing an industry's subsector. It proceeds to evaluate their economic performance before concluding with a discussion on industry trends, considering the innovative aspects of these companies.

5.1. Weaknesses

(I) Strategy: A noticeable shortcoming in strategic thinking was identified among entrepreneurs, as their primary focus was on participating in the Cephalonia winemakers' association and conducting market research in comparison to competitors. These actions, however, do not constitute strategic approaches but rather specific measures. This shortcoming highlights a disconnect between the entrepreneurs' knowledge and skill sets and the core tenets of organizational development theory. Intriguingly, this gap has been previously observed in organizationally weaker micro-firms (Chatzinikolaou & Vlados, 2022), and its persistence in more dynamic micro-firms highlights the need for innovative

partnerships between businesses, academic institutions, and government policies to address this issue (Cai, 2022).

To date, Greek governments have not consistently pursued strategies for promoting product brands within cluster initiatives. This inconsistency has led to the recent proposal of relevant policies. For example, the Institutes of Local Development-Innovation (ILDI), as suggested by Chatzinikolaou and Vlados (2022) aims to support companies in strengthening their strategic, technological, and managerial capabilities. Policies like the ILDI serve as a "business clinic" that fosters collaboration and drives innovation by facilitating effective public-private partnerships.

(II) Technology: The typical confusion about the definition of technology in micro-firms was also confirmed. Several times during interactions with entrepreneurs, there was a reference to machinery and physical equipment as a way to identify specific technological improvements (see technology definition in Figure 2). However, a strong interest and frequent references to knowledge and continuous learning dimensions were observed, giving the impression that these companies tend to make their technological potential more flexible.

In line with the observations on knowledge and continuous learning in micro-firms, Ray Chaudhury et al. (2014) concluded that winemaking requires a continuous learning commitment and that winemakers located in an emerging wine region exhibited gregarious tenacity to learn from mistakes to ensure business success. This reinforces the notion that these companies strive to make their technological potential more flexible by focusing on knowledge acquisition and adaptability.

(III) Management: The common problems of informality and family character in most micro-firms were encountered. This relative organizational inefficiency seemed to be the central problem for these enterprises, which appears to threaten their

competitiveness. As some entrepreneurs mentioned in the interviews, recruiting qualified management staff could help address this weakness —perhaps these firms should turn to the international labor market if the domestic one does not cover their needs. This proposal seems to be in line with Duarte Alonso and Bressan (2015), who studied small wineries and concluded that diversification and extended outward orientation could help improve the performance and resilience of these firms.

(IV) Innovativeness: The pandemic has seemingly induced a sense of insecurity among entrepreneurs, prompting them to confront the rapid pace of change required for business transformation. Interviews have served to heighten their awareness and urgency. To innovate and navigate today's escalating crises, firms that have weathered the pandemic must now implement change management strategies (Chatzinikolaou & Vlados, 2022). This perspective is supported by Wittwer and Anderson (2021), who discuss the impact of COVID-19 on the wine industry. They conclude that wine-exporting countries will face greater challenges than importers, as they must lower prices in the short and medium term. Their findings also show that small premium wine producers have suffered the most, with large producers dominating sales of low-quality wine at reduced prices. However, Solanelles et al. (2021) argue that the pandemic crisis may offer an opportunity to capitalize on anxiolytic wine consumption, suggesting that businesses should consider incorporating this aspect into their missions.

5.2. Strengths

(I) Strategy: Previous research has shown that micro-firms with less competitiveness struggle to articulate their mission and vision within an integrated philosophy (Chatzinikolaou & Vlados, 2022). On the other hand, Robola-producing companies have demonstrated to some extent that they effectively address these strategic abstraction

dimensions, supporting Jordan et al. (2007) that highly-developed wineries perform better on strategic processing and export orientation dimensions to some extent. Additionally, Alonso and Northcote (2010) found that small winery operators in a dynamic wine producing region vary in their understanding and implementation of business vision, stressing the importance of having a clear business vision in determining the survival, access to commercial opportunities, and capitalization of these opportunities. This highlights the significance of a clear business vision and strategic planning for the success and competitiveness of micro-firms, including wineries.

(II) Technology: Skills and knowledge of the subject matter were other important characteristics identified in the employees of these companies. Therefore, their technology appeared to be sufficiently specialized, as their product requires detailed work. This is consistent with findings from previous studies that have emphasized the importance of specialized knowledge and skills in driving innovation and productivity in the wine industry (Meraz-Ruiz et al., 2023). In the context of Munroe's (2022) study, understanding and applying the Vineyard Geological Identity concept also requires expertise in various domains, such as geology, soil science, and viticulture, highlighting the need for specialized knowledge and technology in the industry.

(III) Management: These individuals expressed their desire to make the business more agile, self-rating relevant dimensions with low scores. However, it was found that these negative judgments arise from their ambition to improve their competitiveness. This ambition has been observed in various studies, highlighting the importance of managerial capabilities in achieving competitive advantage (Lorenzo et al., 2018). In fact, Villanueva and Ferrer (2020) found that the managerial skills of winery owners in a new wine region, such as efficiently coordinating resources and adapting to market opportunities, played a

more significant role in their performance and competitive advantage than the differentiation strategy itself.

(IV) Innovativeness: It was observed that the negative conclusions on the "comprehensiveness" criterion resulted from the desire of these entrepreneurs to improve themselves. Also, the frequent reference to qualitative dimensions in the self-assessments of innovativeness demonstrated that these firms, to some extent, effectively synthesize the Stra.Tech.Man dimensions.

The finding that entrepreneurs in Robola-producing companies are eager to improve themselves and effectively integrate the Stra.Tech.Man dimensions aligns with Mehta and Bhanja's (2017) emphasis on the importance of innovation in the wine industry, especially in product development, marketing, and distribution. However, this study also complements the findings of Sánchez-García et al. (2023), who highlighted that the focus on innovation in the wine industry often tends to be too narrow, concentrating primarily on production-related aspects and neglecting the broader organizational context. In this study, the entrepreneurs appear to demonstrate a more holistic understanding of innovation, incorporating not only strategy, technology, and management aspects but also recognizing the crucial role of green transformational leadership, green motivation, and green creativity in driving environmentally sustainable innovations in the wine industry.

5.3. Financial reality check: Examining the accuracy of wineries' post-pandemic performance evaluations

Due to the unavailability of published financial statements for Wineries 1 and 5, and limited data for Winery 3 (2017-2019), reliance on their self-assessments was necessary. It is crucial to consider potential discrepancies between evaluations and actual financial performance. A tendency to overestimate financial performance was noted in Wineries 2

and 4, possibly resulting from subjective Likert-type questions, cognitive biases, limited financial literacy, and focus on other business aspects (Creswell, 2014). Such factors could also influence the self-assessments of Wineries 1, 3, and 5. Specifically, Wineries 1 and 5 reported insufficient resources and liquidity for increased exports and recovering lost ground in the competitive market.

Discrepancies between entrepreneurs' self-assessments and actual financial ratios are evident, as illustrated in Figure 7. For instance, Winery 2 recorded a notable improvement in asset turnover from 1 to 2, but the actual ratio only increased from 0.5 to 0.58 between 2020 and 2021, a modest 16% improvement.

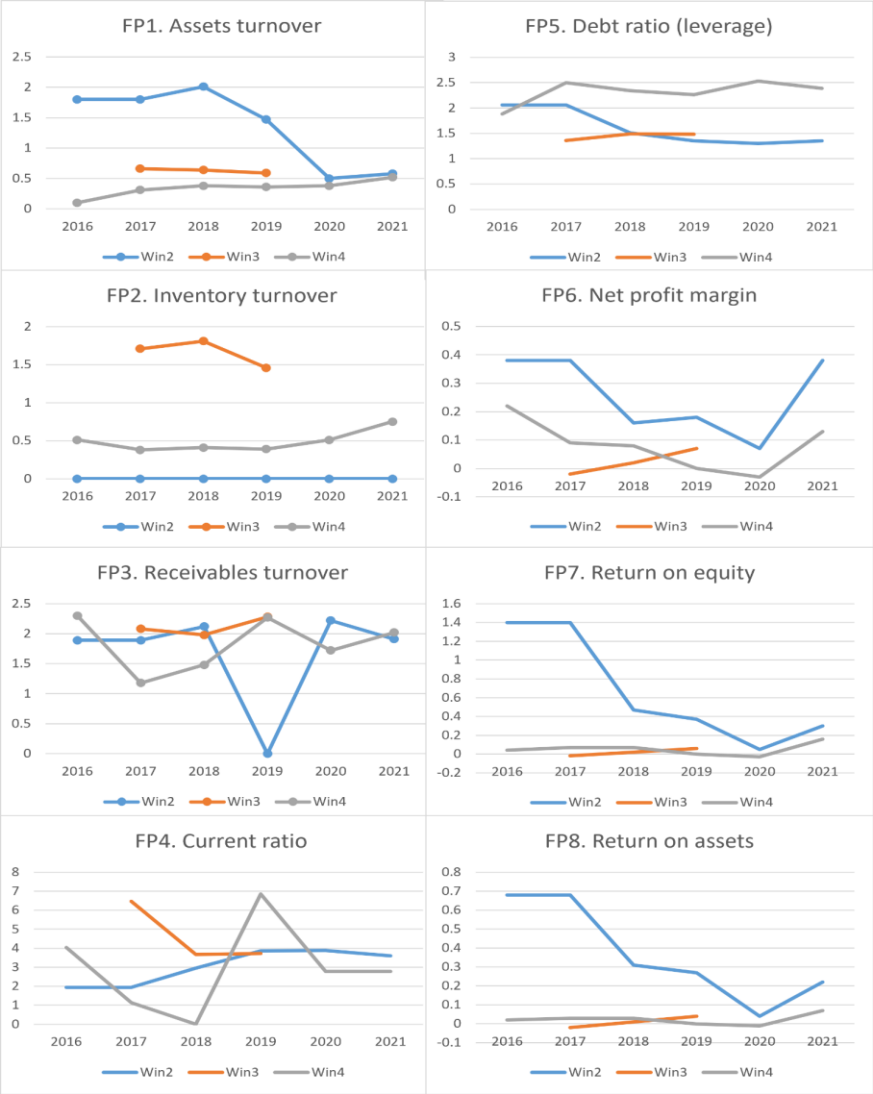


Figure 7: Comprehensive overview of financial performance indicators: Analysis of actual variables

Similar discrepancies were observed in other ratios, such as inventory turnover. Winery 2 recorded no change, while Winery 3 experienced a 15% pre-pandemic drop (2017-2019) and Winery 4 saw a 47% improvement (2016-2021). This improvement could be attributed to increased stock levels across all producers during the pandemic.

In receivables turnover, Winery 2 displayed a slight post-pandemic decrease, contrary to their self-assessment, while Winery 4 improved by 17% (2020-2021). Respondents attribute this to their premium products and loyal customer base. The current ratio showed mixed results, with a minor decline for Winery 2 (2020-2021) and a sharp 60% drop for Winery 4 between 2019 and 2021. A possible explanation is that these firms have adequate resources to cover short-term liabilities. However, it is uncertain whether they are investing their capital effectively or if some liquidity remains inactive. Debt ratio remained relatively stable, contrasting with self-assessments suggesting significant improvements. This indicates that these firms are relatively resilient, as the pandemic did not cause any notable upward trend signifying financing issues.

Profitability indicators, such as net profit margin, increased substantially for Wineries 2 and 4 between 2020 and 2021, reaching 443% and 533%, respectively. Winery 4, in particular, transitioned from negative to positive profitability. Return on equity and return on assets also improved, though actual improvements were generally smaller than self-assessments indicated. Respondents explained that their businesses were impacted by the pandemic, but have since recovered considerably. The improved profitability and high scores on self-assessed resilience through innovativeness seem to support their claims of post-pandemic resilience.

In conclusion, while actual financial ratios do show post-pandemic improvements, they are generally not as pronounced as entrepreneurs' self-assessments suggest. This discrepancy underlines the importance of accurately assessing a business's financial performance to make well-informed decisions and set realistic expectations. Entrepreneurs should recognize and address potential biases and limitations in their self-assessments, ensuring they have a clear understanding of their businesses' true performance.

5.4. Where is this regional wine brand heading?

The literature suggests that building a wine brand with local characteristics is multi-level, involving different parties with converging and diverging interests (Johnson & Bruwer, 2007). It was observed that those firms producing Robola are particularly active in networking with each other in recent years, having formed a related association. Also, one of the firms is a cooperative of smaller producers, which reveals the trend of increased inter-cooperation among them, which seems to be intensifying in the post-COVID-19 era. Several respondents also expressed dissatisfaction with the slow moves to get their region and product a regional brand name that would allow them to increase their exports. They also mentioned that a more sophisticated commercial strategy for the region's wineries would improve them innovatively, addressing some chronic problems they have faced in licensing the number of vineyards they are allowed to grow. The picture extracted from all the self-assessments and open responses is that Robola resembles other clustering forms.

In Porter's (1998, 2000) fundamental contribution, a cluster consists of a broad chain of knowledge, capital-intensive interconnected industries, and geographical concentrations of coevolving firms and institutions. According to respondents, the Robola

cluster appears undercapitalized as significant large-scale investments have not been observed in recent years. Several self-assessments indicated that these businesses are boutique wineries, emphasizing extroversion as they target a niche clientele. In summary, the companies producing this Robola variety are probably at the core of a cluster in its infancy, appearing willing to move in the coming years to further network with each other, gradually building a more sophisticated wine brand.

6. Conclusion

This study explored micro-firms in Cephalonia producing and marketing Robola PDO wine. Utilizing the Stra.Tech.Man Scorecard, it examined how these businesses' owners and managers self-assessed their evolutionary adaptation efforts before and during the COVID-19 pandemic (from 2017 to 2021). The findings offer insights into the entrepreneurial physiological evolution within Cephalonia's wine sector, specifically the PDO Robola variety, and provide predictions for the future development of this emerging cluster.

A) The investigation identified several driving forces behind the transformation of the examined firms' physiology during the study period. The primary factor was the significant disruption caused by COVID-19 in the wine industry's consumption and production. Additionally, a relative lack of extroversion and limited profitability prompted the need for strategic, technological, and managerial changes within these firms.

B) Employing the Stra.Tech.Man approach, this research found that executives in these firms believe they have successfully evolved their organizations' physiology over the examined period. They tended to self-assess their firms as flexible, although with some monad-centeredness elements. The most notable improvements were observed in

strategy, as these companies cultivated their vision and continuously increased exports. Technology appeared to lag in networking conditions, partly explaining the Robola cluster's limited dynamism despite its growth potential. In management, some obstacles were noted, but there was also a disposition towards increased inclusiveness, as the staff —excluding auxiliary workers— were small in number but skilled. Moreover, the frequent reference to qualitative dimensions in self-assessments of innovativeness showed these firms effectively synthesized the Stra.Tech.Man dimensions to some extent.

C) The Robola cluster is characterized by relatively high specialization but limited capital investment. PDO products generally have clustering potential as they necessitate knowledge-intensive business activities. According to Figure 4, a "new star" could emerge in this sector if producers follow directions to create a sophisticated regional wine brand, similar to other successful models abroad (e.g., Atkin et al., 2017; Johnson & Bruwer, 2007).

This study has some limitations, which present opportunities for future research. While the Stra.Tech.Man Scorecard allows self-assessment over two horizons (pre-COVID-19 until the present), longitudinal monitoring and revisiting the firms would further validate the findings. Conducting similar research in other spatial clusters of the wine industry and beyond could offer comparative performance insights, leading to more robust conclusions.

In conclusion, this analysis offers insights into both the theoretical and practical aspects of the Robola variety and similar entrepreneurial ecosystems. From a theoretical standpoint, it deepens our understanding of the Stra.Tech.Man approach's role in shaping entrepreneurial dynamics and business physiology across various sectors and regions. On a practical level, the findings underscore the importance of local and national government

initiatives in promoting these product brands collectively, benefiting both the local community and producers.

While the entrepreneurs' desire to improve through innovation is evident, their financial results reveal relatively limited investment, as well as moderate gains in competitiveness and profitability. It is incumbent upon these agile entrepreneurs to pursue novel avenues for innovation and enhance extroversion, ultimately elevating the variety they produce to the status of a European brand and, potentially, a global one.

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Disclosure statement

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